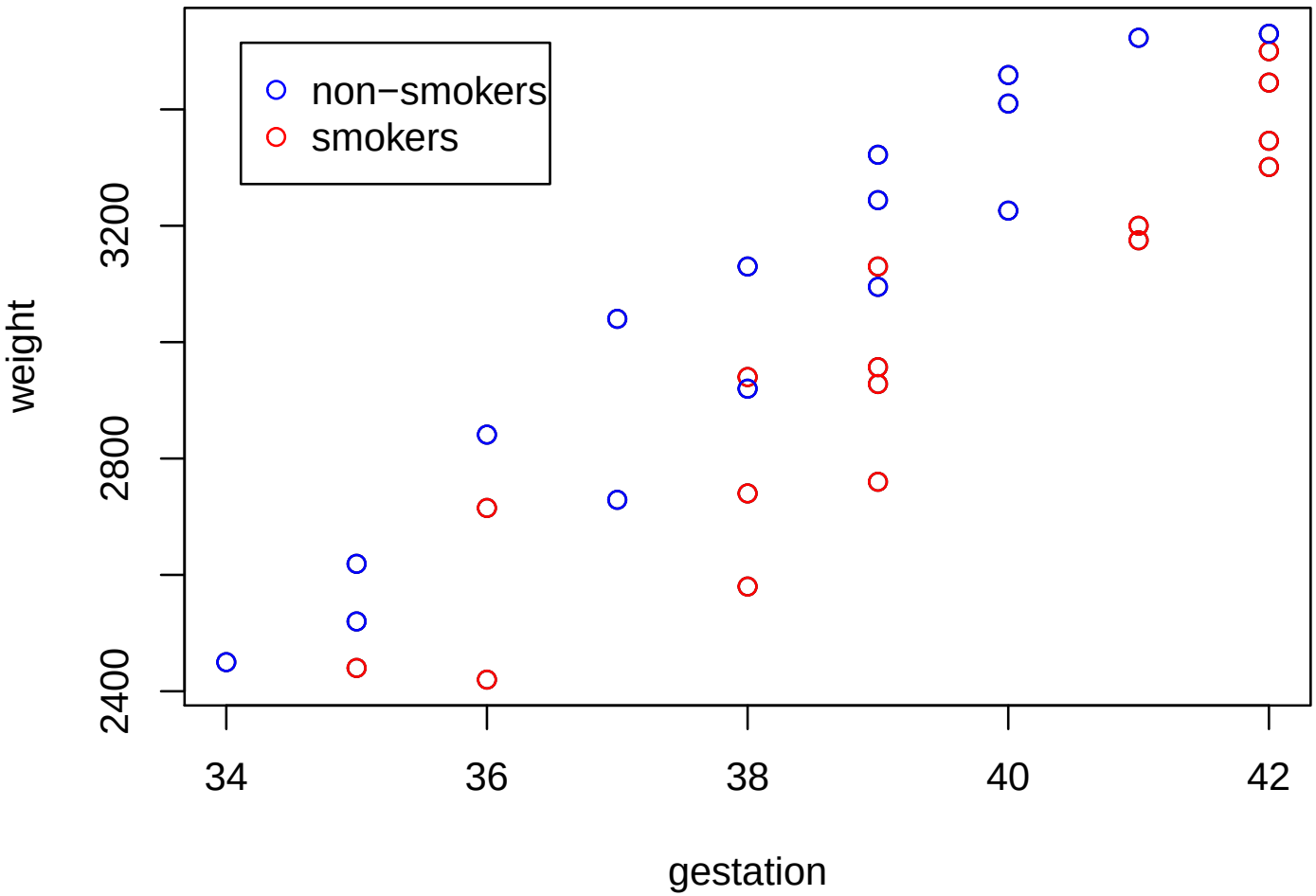
Regression and Predictive Analytics

Stat 3502, Fall 2024

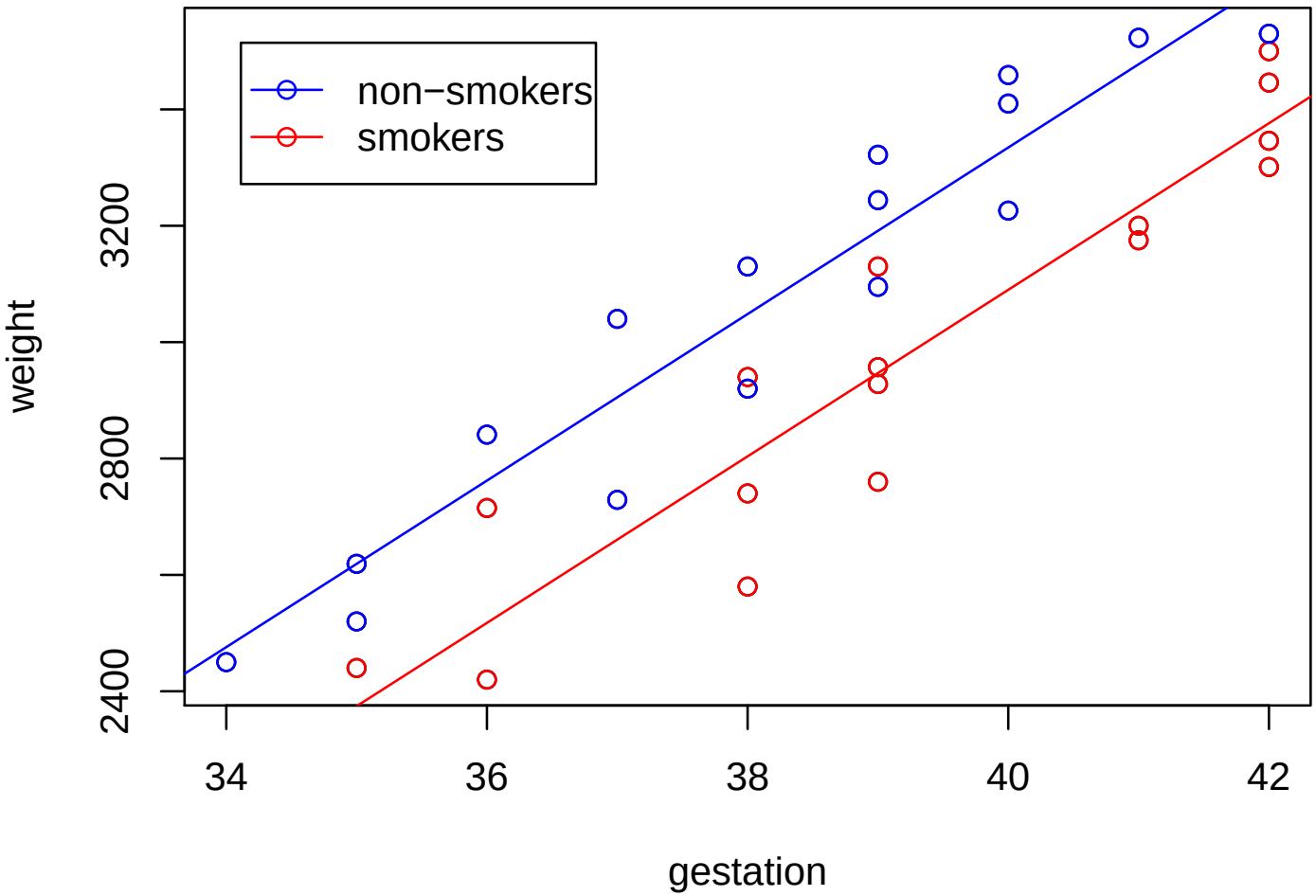
Categorical predictor example

Eg: Use birth.txt for the data.
n = 32 births
Response y: birth weight of baby (in gram)
Predictor x_1: length of gestation (in weeks)
Predictor x_2: smoking status of mother (yes=1, no=0)
Refer to birth1.R

Birth weight scatterplot with smoking status



Birth weight least squares fit



Type of variables

- Continuous response y
- One continuous predictor x_1
- One binary predictor x_2
- x_2 is coded as 1 (yes) or 0 (no)
- x_2 is known as the indicator variable

Population level model

- $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$, where $\varepsilon \sim N(0, \sigma^2)$
- How to interpret β_0, β_1 and β_2?

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- The population has two groups, "yes group" and "no group"
- For the "no group", plug in $x_2 = 0$ and we get

$$y = \beta_0 + \beta_1 x_1 + \varepsilon$$

- Thus β_0 is the intercept for the no group, and β_1 is the slope of x_1 for the no group

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- How to interpret β_0, β_1 and β_2?
- The population has two groups, "yes group" and "no group"
- For the "yes group", plug in $x_2 = 1$ and we get

$$y = (\beta_0 + \beta_2) + \beta_1 x_1 + \varepsilon$$

- Thus $\beta_0 + \beta_2$ is the intercept for the yes group, and β_1 is the slope of x_1 for the yes group

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- β_2 is the change of the intercept from the no group to the yes group

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- x_2 is coded as 1 (yes) or 0 (no)
- Recall $E(\varepsilon) = 0$. Thus

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

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- β_1 is the change of units in the mean response $E(y)$ for each unit increase in the continuous predictor x_1, regardless of the group
- β_2 is how much higher (or lower) the mean response $E(y)$ of the yes group is than the that of the no group for any value of x_1

Birth weight example

- The fitted model is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 = -2389.57 + 143.10x_1 - 244.54x_2$$

- For the non-smokers, $x_2 = 0$ and the fitted regression line is

$$\hat{y} = -2389.57 + 143.10x_1$$

- For the smokers, $x_2 = 1$ and the fitted regression line is

$$\hat{y} = -2389.57 + 143.10x_1 - 244.54 = -2634.11 + 143.10x_1$$

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- The fitted regression line for non-smokers are above the fitted regression line for smokers. Why?

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- A: Conduct hypothesis testing $H_0 : \beta_2 = 0$ v.s. $H_0 : \beta_2 \neq 0$. From the t-test, we see that p-value is smaller than 0.05, and conclude that the effect of smoking on birth weight is significant after taking into account of gestation length. We can also use the extra sums of squares F-test to get the same conclusion.

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- Q: Can we use the extra sums of squares F-test?
- A: We can not use the F-test, as the alternative hypothesis of F-test has to be two-sided.

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Why not fit two separate models, one for the smokers group, and one for the non-smokers group?

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- Do we prefer smaller standard error for $\hat{\beta}_1$?

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- From the joint model, the confidence interval is $(2740.60, 2866.79)$
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- We prefer shorter confidence intervals from the joint model (126.2) over the separate model (153.6)

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- If we fit two separate models for smokers and non-smokers, we can not easily answer this question.
- Simply test $H_0 : \beta_2 = 0$ v.s. $H_0 : \beta_2 \neq 0$ in the joint model.

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- What happens if we fit regression model with both x_2 and x_3?
- One of them will be automatically dropped
- We only need one indicator variable for a binary predictor .
- In general, $k - 1$ indicator variables are needed for a categorical predictor with k categories

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- What if we code $x_2 = 1$ for smoking and $x_2 = -1$ for non-smoking? Will the fitted regression lines change?
- The estimated regression coefficient $\hat{\beta}_1$ is the same in two different coding schemes
- The estimated intercept and the estimated slope for the binary variable differ in the two coding schemes
- There will be no changes to the final fitted regress lines
- Refer to birth3.R

Interpretation of birth weight example, again

Fitted model is $\hat{y} = -2389.57 + 143.10x_1 - 244.54x_2$

- Q: does the effect of the gestation length on mean birth weight depend on whether or not the mother is a smoker?
- A: No. Regardless of whether or not the mother is a smoker, for each additional one-week of gestation, the mean birth weight is predicted to increase by 143.1 grams

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- Q: does the effect of smoking on mean birth weight depend on the length of gestation?
- A: No. For a fixed length of gestation, the mean birth weight of babies born to smoking mothers is predicted to be 244.54 grams lower than the mean birth weight of babies born to non-smoking mothers.

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- A: No. For a fixed length of gestation, the mean birth weight of babies born to smoking mothers is predicted to be 244.54 grams lower than the mean birth weight of babies born to non-smoking mothers.
- The lack of interaction between the two predictors is exhibited by the parallelness of the two lines

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- $x_1 x_2$ is known as the interaction between x_1 and x_2

Population level interaction model

- $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \varepsilon$
- For the "no group", plug in $x_2 = 0$ and we get

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- Thus β_0 is the intercept for the no group, and β_1 is the slope of x_1 for the no group

Population level interaction model

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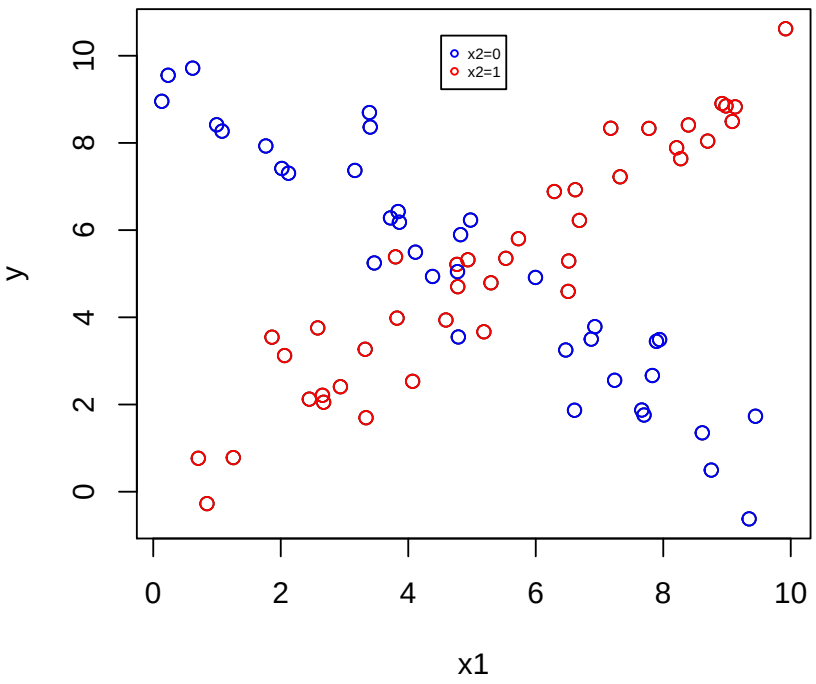
- For the "yes group", plug in $x_2 = 1$ and we get

$$y = (\beta_0 + \beta_2) + (\beta_1 + \beta_3) x_1 + \varepsilon$$

- Thus β_2 is the change of intercept from the no group to the yes group, and β_3 is the change of the x_1 slope from the no group to the yes group

Toy example



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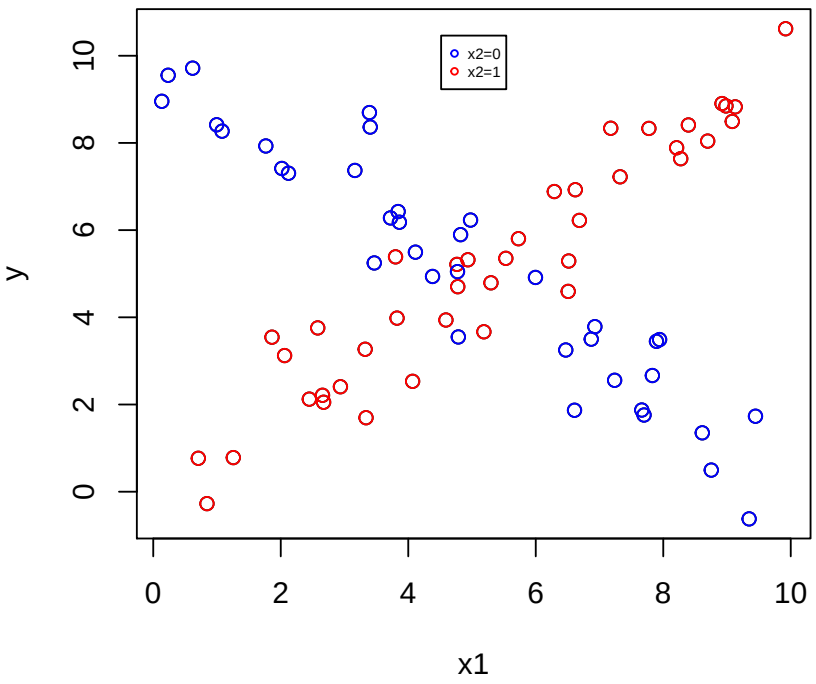


Refer to interaction.R

$x_1 \sim Unif(0, 10)$

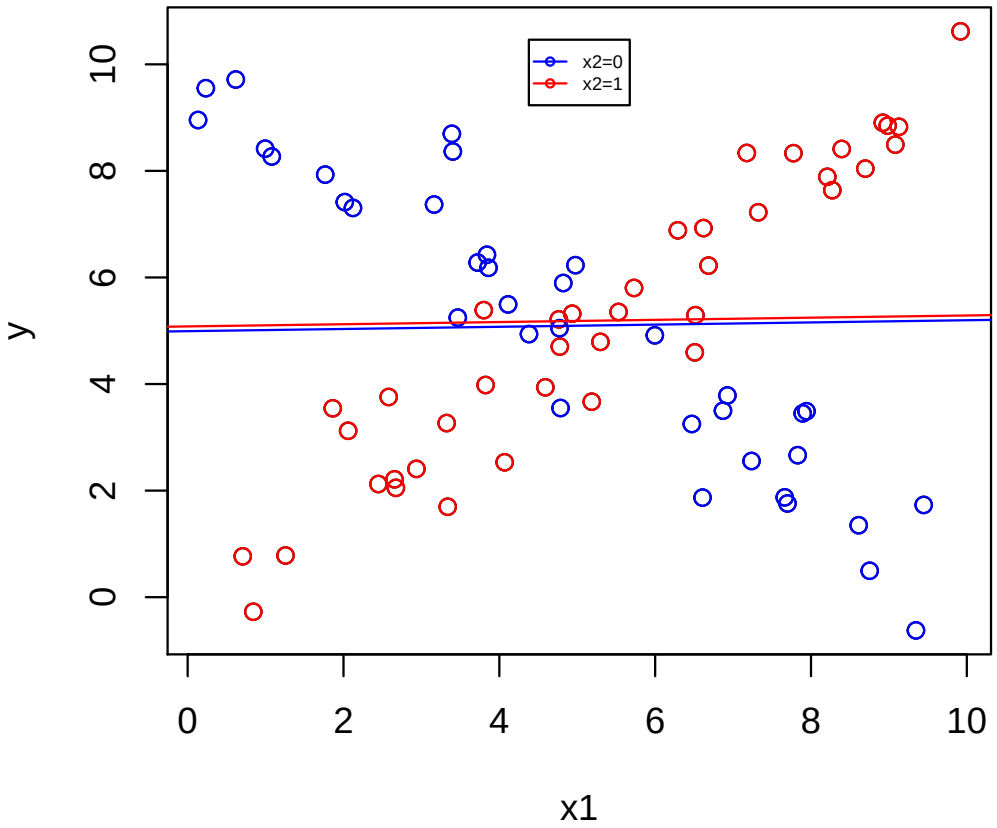
$x_2 \sim Bernoulli(0.5)$

$\varepsilon \sim N(0, 1)$

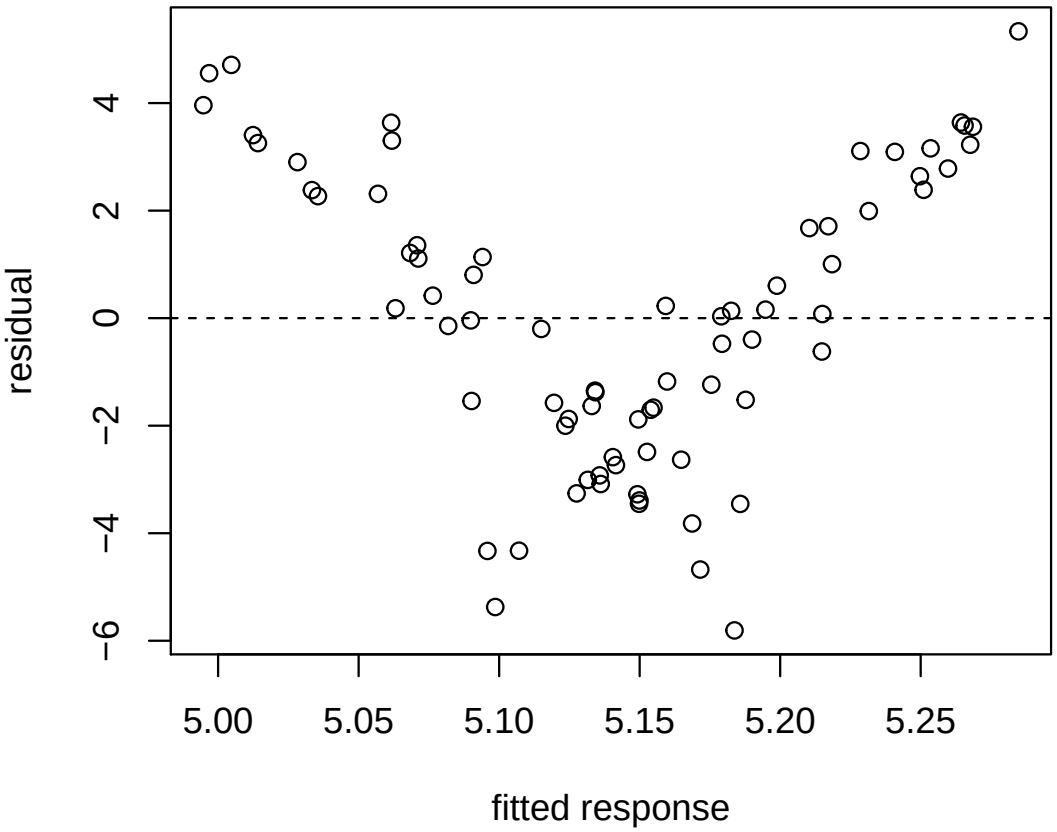
True model is

$$y = 10 - x_1 - 10x_2 + 2x_1x_2 + \varepsilon$$

Fit the data without interaction



Residual v.s. fitted response without interaction



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- If we fit the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$, the t-tests will suggest that neither β_1 (p-value 0.865) or β_2 (p-value 0.889) is significantly different from 0

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- Is y related to x_2 (group effect)? No according to the p-value. Yes according to the scatter plot

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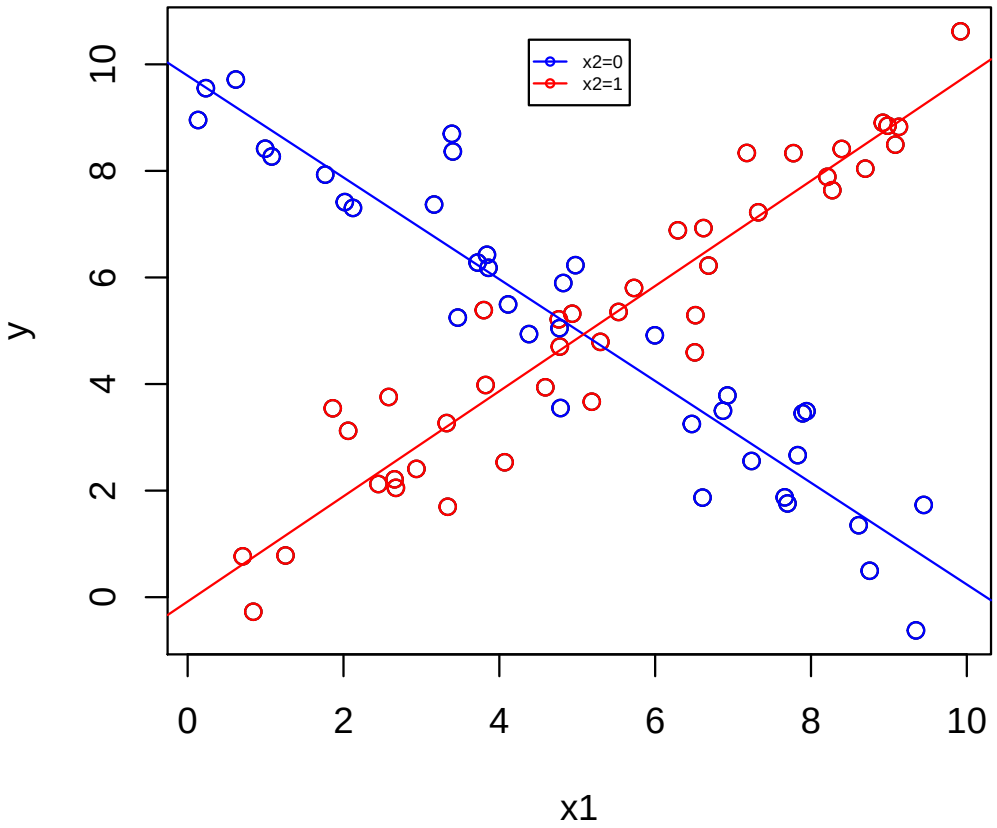
- Is y related to x_1 (the continuous predictor)? No according to the p-value. Yes according to the scatter plot
- Is y related to x_2 (group effect)? No according to the p-value. Yes according to the scatter plot
- Significant trend in the residual plot suggests the model does not fit well

Fit the data with interaction



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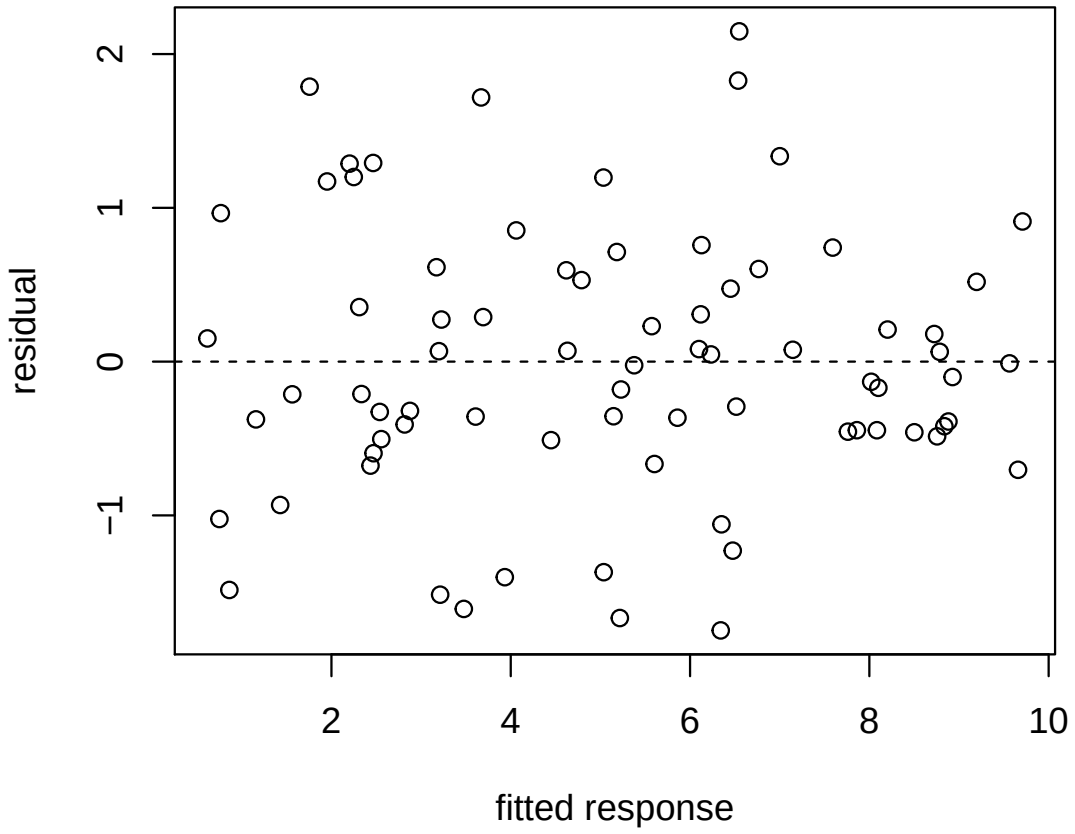

Toy example with interaction

- The fitted regression lines for two group cross (interact)
- The slopes of the two fitted regression lines are different
- The p-values for all the t-tests are significant (not shown here)
- The residual plot no longer has any trend, suggesting the interaction model fits the data well

Toy example with interaction

- Does the effect of x_1 on response depend on the group variable x_2?
- Yes according to p-value from t-test $H_0 : \beta_3 = 0$ v.s. $H_0 : \beta_3 \neq 0$

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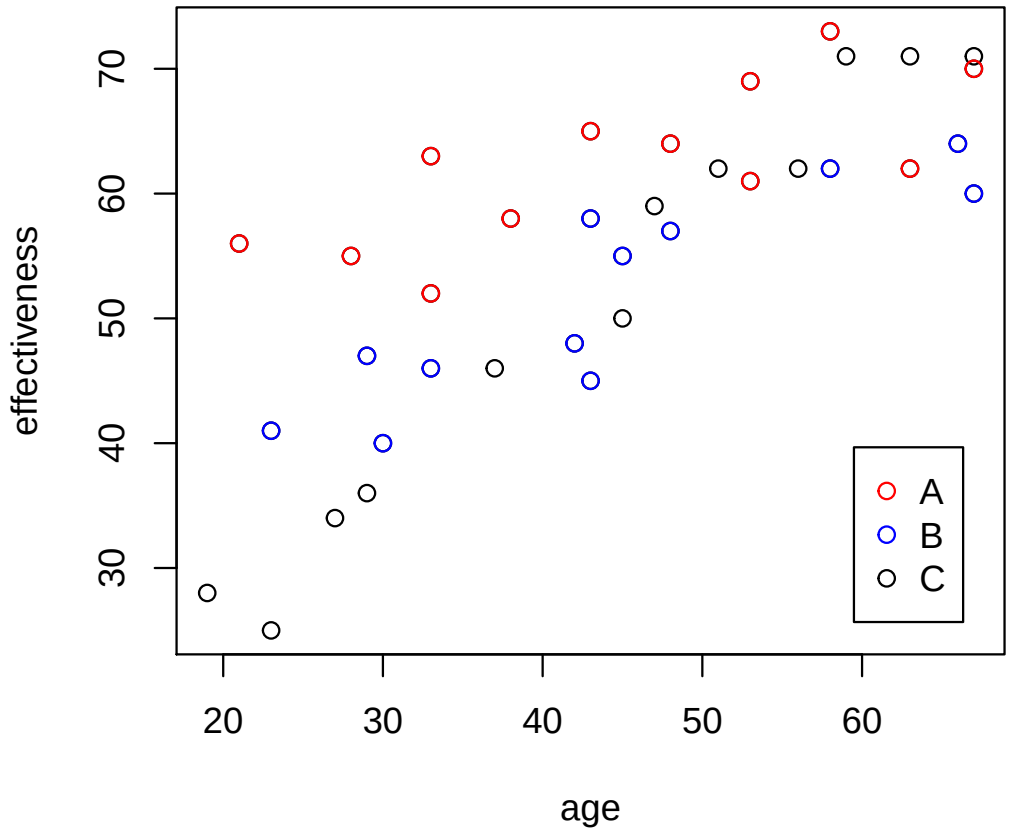
- Yes according to p-value from t-test $H_0 : \beta_3 = 0$ v.s. $H_0 : \beta_3 \neq 0$
- Is y related to x_1 (the continuous predictor)?
- Question is not very important as the effect of x_1 on y depends on x_2. Should not talk about effect of x_1 on y without considering x_2
- Do not interpret a main effect in the presence of an interaction. Do not test $H_0 : \beta_1 = 0$ v.s. $H_a : \beta_1 \neq 0$ if already have β_3 significantly different from 0.

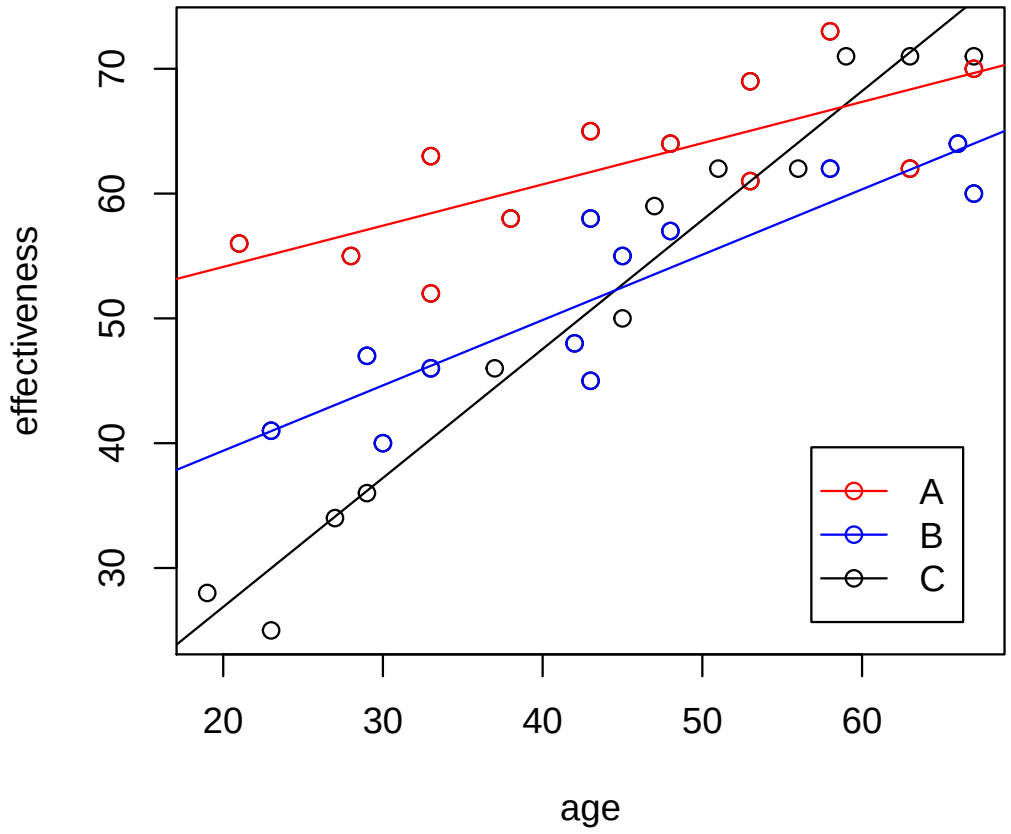
Predictor with 3 categories

Eg: Use depression.txt for the data.
n = 36 patients who receive treatment A, B or C for depression
Response y: effectiveness of the treatment
Predictor x_1: age of patient
Predictor x_2: indicator for treatment A ($x_2 = 1$ means getting treatment A, otherwise $x_2 = 0$)
Predictor x_3: indicator for treatment B ($x_3 = 1$ means getting treatment B, otherwise $x_3 = 0$)
Refer to interaction_depression.R

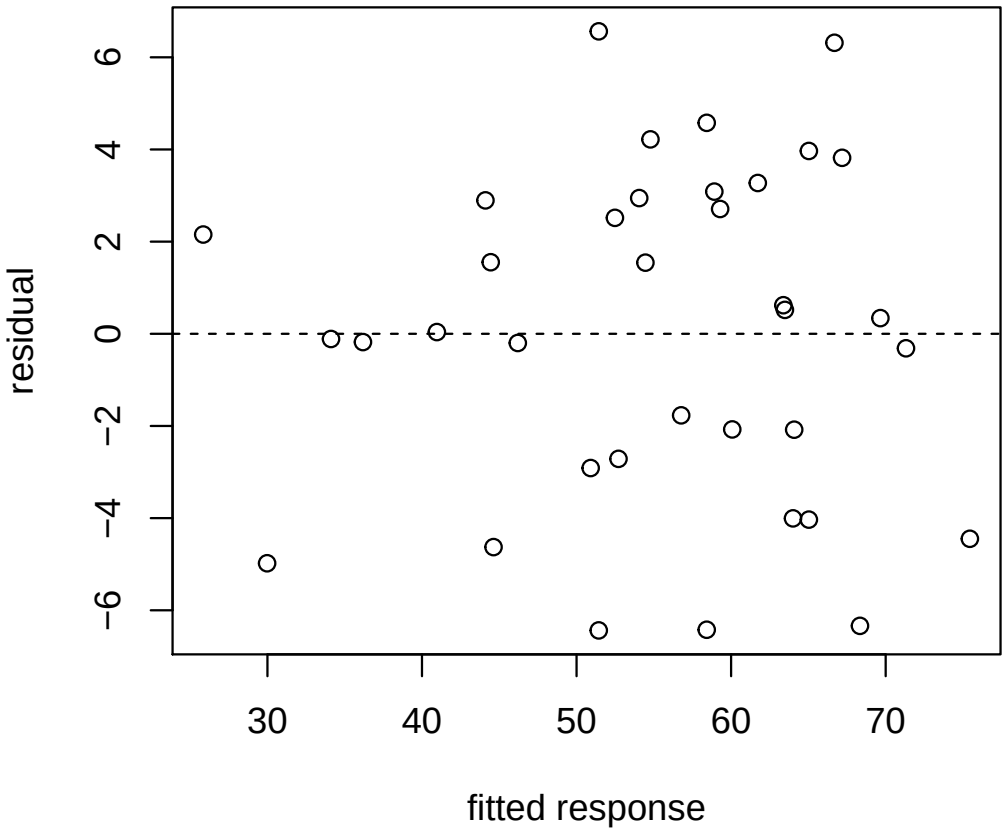
Scatterplot for depression data



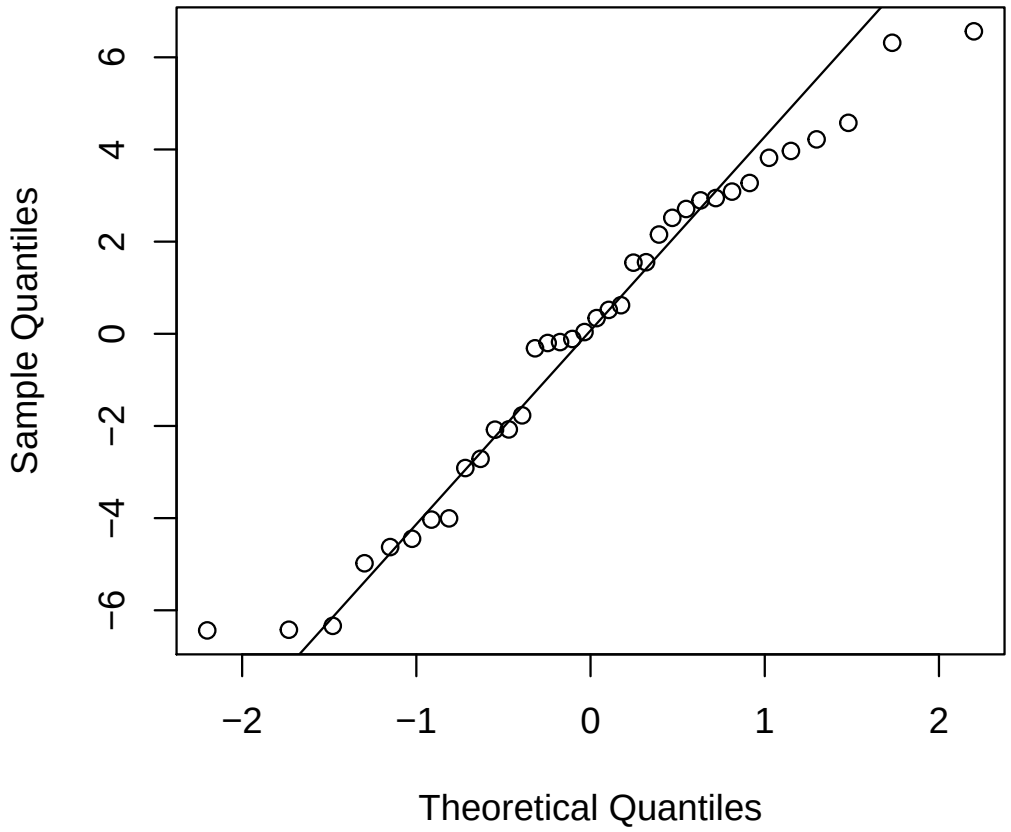
Fitted regression lines with interaction



Residual v.s. fitted response for depression data



Residual QQ plot for depression data



Population level model for depression data

- $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_3 + \varepsilon$
- For treatment C group, plug in $x_2 = 0$ and $x_3 = 0$

$$y = \beta_0 + \beta_1 x_1 + \varepsilon$$

- β_0 is the intercept for treatment C group, and β_1 is the slope for treatment C group

Population level model for depression data

- $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_3 + \varepsilon$
- For treatment A group, plug in $x_2 = 1$ and $x_3 = 0$

$$y = (\beta_0 + \beta_2) + (\beta_1 + \beta_4) x_1 + \varepsilon$$

- β_2 is the change of intercept from treatment C group to the treatment A group, and β_4 is the change of slope from treatment C group to the treatment A group

Population level model for depression data

- $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_3 + \varepsilon$
- For treatment B group, plug in $x_2 = 0$ and $x_3 = 1$

$$y = (\beta_0 + \beta_3) + (\beta_1 + \beta_5) x_1 + \varepsilon$$

- β_3 is the change of intercept from treatment C group to the treatment B group, and β_5 is the change of slope from treatment C group to the treatment B group